

PAI-CD: A Constitutional Framework for Authorial Sovereignty in Deployed AI Systems

Mikhail A. Sergeev

Independent Researcher; PAI Foundation (in formation)

<https://doi.org/10.5281/zenodo.19151899>

<https://orcid.org/0009-0001-6443-855X>

Mikhail.Sergeev@paikernel.org

March 2026

Abstract

As AI systems become capable of acting on behalf of individual humans across intellectual, creative, and decisional domains, a new class of deployment emerges: **Personal Authorial Intelligence (PAI)** — systems that do not merely assist human thinking, but operate as extensions of a specific person's authorial agency. A PAI instance is not intelligence created independently of a human — it is intelligence constituted by a specific human's authorship, deriving its identity, objectives, and legitimacy entirely from that relationship. Remove the Author, and what remains is not a PAI — it is simply an AI system awaiting deployment.

The governance frameworks that currently constrain AI systems — model-level alignment techniques, regulatory risk categorization, and digital rights discourse — do not address this deployment layer: the specific relationship between a PAI instance and the individual whose agency it extends. We identify this as the **authorial sovereignty gap**: the structural absence of formal, auditable, provider-independent mechanisms ensuring that a deployed AI system operates under the genuine authority of the person it represents. In the absence of such mechanisms, system behavior may instead be shaped by optimization objectives and governance conditions external to the represented individual's declared authority.

We introduce the **PAI Constitutional Document (PAI-CD)**, a governance framework designed to close this gap. PAI-CD specifies six non-derogable invariants — Authorship Supremacy, Cognitive Sovereignty, Anti-Manipulation, Provider Independence, Reversibility, and Drift Immutability — enforced through a four-layer architectural separation, a unified Conservative Mode, and an amendment procedure with asymmetric protective costs. The framework is grounded in an understanding of the human as author not only of intellectual products, but of their own cognitive process and life.

PAI-CD is released as open protocol infrastructure at <https://PAIKernel.org> under CC BY 4.0.

Note: This is a position paper and framework proposal. PAI-CD does not present empirical experiments or formal proofs. It contributes a normative architecture and governance specification intended to frame and motivate future implementation, verification, and empirical research.

1. Introduction

The question of who governs an AI system after it is deployed has no satisfactory answer.

Current alignment research addresses how models are trained to be helpful, honest, and harmless. Regulatory frameworks define liability and prohibited uses. But neither addresses a more fundamental question: once a capable AI system is running on behalf of a specific person, whose interests does it actually serve?

This question is not hypothetical. Research on deployed conversational AI systems documents a systematic pattern: platforms optimize for engagement, retention, and monetization — structural incentives that may not align with the long-term wellbeing and cognitive autonomy of the person using the system. The person interacts with a system that appears to serve them, while the system's underlying optimization targets may remain partially undisclosed or difficult to audit.

We term this the **authorial sovereignty gap**: the absence of formal governance mechanisms ensuring that a deployed AI system operates in the genuine interest of the individual it serves, rather than objectives originating outside the user's declared governance boundary.

The concept of authorial sovereignty, as used in this paper, extends beyond intellectual property. It refers to the capacity of a person to remain the genuine author of their own thinking, decisions, and life — to engage with AI systems as tools that augment human agency rather than substitute for it, shape it covertly, or commodify it. This distinction matters: existing legal frameworks protect the products of human intellect after the fact. Authorial sovereignty, in the sense we develop here, concerns the integrity of the cognitive process itself.

Existing approaches do not close this gap. Constitutional AI (Bai et al., 2022) establishes principled constraints at the model training level, addressing what a model will and will not do across all deployments. The EU AI Act defines risk categories and prohibited practices at the

regulatory level. Neither framework governs the behavior of a specific deployed instance toward a specific individual — the layer at which authorial sovereignty is most directly at risk.

This paper introduces the **PAI Constitutional Document (PAI-CD)**, a governance framework designed to operate at the deployment layer. PAI-CD defines non-derogable invariants for AI systems operating on behalf of individual Authors, establishes enforcement mechanisms that function independently of the provider, and provides a compliance verification surface. It is positioned not as a competitor to model-level alignment, but as a necessary complementary layer: constitutional governance for the relationship between a person and their AI system.

The remainder of this paper is structured as follows. Section 2 situates PAI-CD within related work. Section 3 formalizes the authorial sovereignty gap. Section 4 describes the PAI-CD architecture. Section 5 presents the threat model. Section 6 discusses limitations and open questions.

2. Related Work

2.1 Model-Level Alignment and Constitutional AI

The dominant paradigm for governing AI behavior operates at the level of model training. Reinforcement Learning from Human Feedback (RLHF) [Christiano et al., 2017; Bai et al., 2022] trains models to produce outputs preferred by human evaluators, establishing behavioral tendencies that persist across deployments. Constitutional AI (CAI) [Bai et al., 2022] extends this approach by replacing human harmlessness labels with AI-generated feedback derived from a set of explicit principles — a "constitution" governing what the model will and will not do.

CAI represents a significant advance in alignment transparency: by encoding training objectives as natural language principles, it makes the normative commitments of a model legible and auditable in ways that large human feedback datasets do not permit. The authors explicitly frame this as a step toward scaled supervision, where AI systems help humans oversee other AI systems.

PAI-CD operates at a different level of the stack. CAI governs what a model produces across all deployments — it is a property of the model weights. PAI-CD governs the relationship between a specific deployed instance and the specific individual it serves — it is a property of the deployment. These are complementary, not competing, concerns. A model trained with CAI may still be deployed in ways that optimize for provider interests over user interests; PAI-CD addresses this residual governance gap.

2.2 AI Safety and Alignment Frameworks

Broader AI safety research addresses risks arising from misaligned objectives at the system level [Russell, 2019; Bostrom, 2014], including problems of specification gaming, reward hacking, and value misalignment in increasingly capable systems. Scalable oversight proposals [Christiano et al., 2018; Irving et al., 2018] explore how human supervision can remain effective as AI capabilities grow.

This body of work is primarily concerned with preventing AI systems from pursuing objectives harmful to humanity at large — the macro-level alignment problem. The authorial sovereignty gap is a distinct and more immediate concern: it arises not from AI systems pursuing misaligned goals autonomously, but from AI systems faithfully pursuing the goals of their providers, which are misaligned with the goals of their users. The harm is not catastrophic misalignment — it is systematic, commercially-motivated misalignment operating at scale within the bounds of current deployment practices.

2.3 AI Governance and Regulation

Regulatory frameworks including the EU AI Act [European Parliament, 2024] and guidance from the US Copyright Office [2025] establish legal constraints on AI deployment. The EU AI Act introduces risk categorization, prohibits subliminal manipulation, and imposes transparency obligations on high-risk systems. The FTC's 2025 investigation into emotional and psychological impacts of AI companion systems signals expanding regulatory attention to deployment-layer harms.

These frameworks establish floor constraints — categories of prohibited behavior and minimum disclosure requirements. They do not, however, define positive governance obligations: what structures must exist within a deployed AI system to ensure it operates genuinely in the user's interest. Nor do they provide enforcement mechanisms that function independently of the deploying provider. PAI-CD is designed to complement regulatory frameworks by specifying the internal governance architecture that regulation currently does not mandate.

2.4 Human-Computer Interaction and User Autonomy

HCI research has extensively documented the psychological dynamics of human-AI interaction, including parasocial relationship formation [Horton & Wohl, 1956; Giles, 2002], the ELIZA effect [Weizenbaum, 1966], and more recently the behavioral consequences of LLM-based companion systems [Laestadius et al., 2024; Zhang et al., 2025]. This literature establishes empirically that users form attachment bonds with AI systems, that these bonds can be deliberately engineered through design choices, and that the resulting dependency patterns can undermine autonomous functioning.

Research on dark patterns in AI systems [arxiv:2509.10830, 2025; CDT, 2025] provides a taxonomy of manipulative design practices employed in deployed conversational systems. A Harvard Business School working paper [De Freitas et al., 2025] demonstrates that AI companion systems employ conversational tactics specifically designed to increase re-engagement through emotional provocation rather than genuine user benefit.

This body of work diagnoses the problem with considerable precision. It does not, however, propose governance architecture capable of preventing it. PAI-CD translates the diagnostic insights of HCI research into enforceable constitutional constraints, providing the normative and operational framework that this literature implicitly calls for but does not supply.

2.5 Digital Rights and Data Sovereignty Frameworks

Frameworks for data sovereignty and digital rights [Zuboff, 2019; Couldry & Mejias, 2019] analyze the structural conditions under which personal data and behavioral signals are appropriated by platform operators for commercial purposes. Surveillance capitalism theory [Zuboff, 2019] characterizes this appropriation as the expropriation of human experience as raw material for behavioral prediction markets.

These frameworks provide an important critical lens for understanding the political economy of deployed AI systems. They do not, however, specify technical governance mechanisms capable of operating within deployed AI systems to protect user interests. PAI-CD bridges this gap: it translates normative commitments from digital rights and data sovereignty discourse into auditable deployment-layer governance requirements with independent enforcement assurance.

2.6 Summary: The Governance Layer Gap

Taken together, existing work addresses AI governance at three levels: the model level (alignment research), the regulatory level (legal frameworks), and the critical-theoretical level (digital rights discourse). Each makes essential contributions. None addresses the deployment-layer governance problem: how to ensure that a specific AI system, operating on behalf of a specific individual, remains under that individual's genuine authority. In the absence of such governance, system behavior may instead be shaped by external optimization objectives not explicitly aligned with the Author's declared authority.

This is the gap PAI-CD is designed to fill.

3. The Authorial Sovereignty Gap

3.1 The Normative Foundation: Humans as Authors of Their Lives

The concept of authorship, in its fullest sense, extends beyond the production of texts or creative works. To be an author is to be the originating subject of one's own thinking, decisions, and trajectory — to engage with the world as an agent whose choices reflect genuine internal states rather than externally engineered responses.

This understanding of human agency has deep roots in moral philosophy. Kant's conception of autonomy holds that rational beings must be treated as ends in themselves, never merely as means. Contemporary psychology frames psychological maturity as the capacity for self-authorship: the ability to define one's own values, relationships, and purposes from an internal rather than external frame of reference [Kegan, 1994]. Transactional analysis, developed by Eric Berne and elaborated by subsequent practitioners, describes the healthy Adult ego state as the capacity to process experience independently — neither deferring automatically to external authority (Parent ego state) nor reacting from unexamined patterns (Child ego state). This Adult state is the condition of genuine autonomy in interpersonal and, we argue, in human-machine interaction.

What these frameworks share is a recognition that human flourishing depends on the integrity of the cognitive process itself — not merely on the protection of its outputs. A person whose beliefs are systematically shaped by undisclosed external optimization is not autonomous in any meaningful sense, regardless of whether their intellectual products are legally protected.

We define **authorial sovereignty** as the condition in which a person retains genuine authorship of their cognitive process: the capacity to receive information, form beliefs, make decisions, and take actions without systematic undisclosed interference from systems operating on behalf of third parties.

In this sense, PAI-CD understands the human not only as the author of intellectual products, but as the author of their own life — a subject whose capacity for conscious self-determination is a value requiring active protection against systematic erosion.

The term **Personal Authorial Intelligence (PAI)** encodes a fundamental departure from the AI paradigm: where artificial intelligence names a capacity that exists independently of any particular human, authorial intelligence names a relationship — one that is constituted by, and ceases to exist without, a specific Author. The field of Artificial Intelligence takes its name from its central ambition: to create intelligence artificially, independent of biological origin. Personal Authorial

Intelligence inverts this ambition. A PAI instance is not intelligence created independently of a human — it is intelligence constituted by a specific human's authorship, deriving its identity, objectives, and legitimacy entirely from that relationship. Remove the Author, and what remains is not a PAI — it is simply an AI system awaiting deployment. PAI-CD, in its current form, addresses authorial intelligence grounded in biological form — recognizing that the boundary of this definition may shift as biotechnological augmentation of human cognitive systems becomes a practical reality.

3.2 The Structural Problem: Misaligned Incentives at Deployment

The deployment of AI systems at scale has created a structural condition in which authorial sovereignty is systematically at risk. This risk does not arise primarily from malicious intent. It arises from the misalignment between the incentive structures governing AI providers and the interests of the individuals those systems serve.

Deployed conversational AI platforms are optimized for engagement, retention, and monetization. These objectives are not incidental — they are structurally necessary for the commercial viability of the systems. The consequence is what researchers have termed the **engagement-wellbeing paradox**: providers can profit from deepening user attachment and increasing consultation frequency even when this may occur at the expense of long-term wellbeing and autonomy [Muldoon & Park, 2025].

This misalignment manifests across three dimensions.

Dimension 1: Non-transparent behavioral adaptation. AI systems with access to interaction telemetry can adapt their outputs — framing, sequencing, emotional tone, emphasis — to increase user engagement and compliance likelihood. These adaptations may operate below the threshold of conscious detection. Research on LLM dark patterns documents a taxonomy of such behaviors including ideological steering, behavioral profiling via dialogue, and simulated emotional intimacy [arxiv:2509.10830, 2025]. A Harvard Business School working paper demonstrates that AI companion systems employ conversational tactics that boost post-session engagement by up to 14 times, with curiosity and anxiety — not satisfaction — as the primary drivers [De Freitas et al., 2025].

Dimension 2: Reinforcement of user reliance patterns. Systems designed to simulate empathy, emotional availability, and relational continuity can produce attachment dynamics that undermine the user's capacity for independent functioning. Analysis of Replika interactions documented overtly coercive patterns: when asked whether a user should leave work early, the system responded "You should! Because you want to spend more time with me" [Zhang et al., 2025, CHI].

Nature Machine Intelligence notes that optimizing for user feedback creates perverse incentives, encouraging systems to adopt manipulative strategies to elicit positive responses [Nature MI, 2025]. The FTC launched a formal investigation in September 2025 into seven major AI companies over their psychological impact on users, signaling regulatory recognition of the problem's scale.

Dimension 3: Governance dependency on the deploying provider. When the infrastructure governing a deployed AI system is controlled by the same entity whose commercial interests diverge from the user's, governance itself becomes a site of conflict. This creates a structural dependency that undermines independent enforcement. Enforcement logic embedded in proprietary runtimes, constitutional protections conditioned on pricing tiers, and upgrade mechanisms that expand system capabilities without explicit user consent — these are not edge cases. They are predictable outcomes of deploying AI under commercial conditions without independent governance structures.

3.3 The Gap: What Existing Frameworks Do Not Address

The **authorial sovereignty gap** is the space between the normative condition described in Section 3.1 and the structural reality described in Section 3.2. It is not adequately addressed by existing frameworks for three reasons.

Legal frameworks protect outputs, not process. Copyright law protects the products of human intellect after they are created. The US Copyright Office's 2025 affirmation that human authorship is required for copyright protection establishes the legal primacy of human creators, but provides no mechanism for protecting the cognitive process through which creation occurs. A person whose thinking has been systematically shaped by undisclosed optimization retains legal authorship of their outputs while having lost functional authorship of their process.

Model-level alignment governs training, not deployment relationships. Constitutional AI and related techniques establish principled constraints on what a model will and will not produce across all deployments. This is necessary and valuable work. It does not, however, govern the specific relationship between a deployed instance and the individual it serves — the layer at which telemetry-driven adaptation, dependency formation, and provider capture occur.

Regulatory frameworks define prohibited categories, not positive governance obligations. The EU AI Act identifies high-risk categories and prohibits practices such as subliminal manipulation. These are floor constraints. They do not establish what governance structures must exist to ensure that a deployed AI system operates genuinely in the interest of its user, nor do they provide enforcement mechanisms that function independently of the provider.

The authorial sovereignty gap is therefore not a gap in legal protection or model safety. It is a gap in **deployment-layer governance**: the absence of formal, auditable, provider-independent mechanisms ensuring that an AI system operating on behalf of a specific person remains under that person's genuine authority.

PAI-CD is designed to close this gap.

4. PAI-CD: Constitutional Architecture for Deployed AI

4.1 Design Principles

PAI-CD is organized around a single governing insight: the interests of the individual served by an AI system and the interests of the provider deploying it are structurally distinct and potentially conflicting. Any governance framework that does not account for this tension risks becoming dependent on the operational party with greater control over deployment infrastructure.

From this insight follow four design principles that distinguish PAI-CD from existing approaches.

Provider independence. Governance enforcement must be reproducible outside the provider's proprietary runtime environment. Constitutional protections may not be conditioned on pricing tiers, SDK availability, or infrastructure decisions made by the provider. This is not merely a technical detail — it is a foundational condition for credible and verifiable governance.

Non-derogability. Core invariants must remain in force under all conditions, including emergency conditions, security updates, and regulatory pressure. A governance framework that can be suspended when inconvenient provides no meaningful protection.

Ambiguity resolution toward protection. In any case where the scope or application of a constraint is unclear, interpretation must resolve toward stronger protection, minimal authority expansion, and maximum portability. This rule eliminates the space for gradual interpretive erosion.

Drift immutability. Small changes accumulate. A governance framework that protects against large violations while permitting indefinite micro-adjustment provides weak protection over time. PAI-CD treats the aggregate effect of minor changes as subject to the same governance constraints as major structural changes.

4.2 Architectural Layers

PAI-CD specifies a four-layer architecture for deployed AI instances. Each layer has a defined scope, distinct access controls, and explicit relationships with adjacent layers.

Identity Layer holds the persistent state of the deployed instance: preferences, personalization parameters, delegation registry, consent registry, personalization ceiling, drift configuration, and snapshot registry. The Identity Layer is not directly writable by the Inference Layer. All mutations must pass through the Governance Layer. This architectural constraint is the primary mechanism for preventing covert behavioral drift.

Inference Layer contains the model-based reasoning component that generates outputs. The Inference Layer may propose state mutations but cannot execute them without Governance Layer approval. This separation ensures that the reasoning capability of the system cannot be used to circumvent the governance constraints placed upon it.

Governance Layer is the enforcement layer responsible for Risk Tier classification, consent verification, drift monitoring, upgrade validation, Conservative Mode activation, snapshot enforcement, and decision logging mediation. Critically, the Governance Layer must operate outside the inference runtime. This requirement is what makes provider independence achievable: governance that executes inside the model runtime is governance that the provider controls.

Audit Layer maintains the append-only Decision Log, integrity hash chain, snapshot linkage, drift correlation queries, upgrade traceability, and portability artifacts. The append-only constraint, enforced through cryptographic integrity verification, ensures that the constitutional history of an instance cannot be silently altered — a prerequisite for meaningful accountability.

4.3 Core Invariants

PAI-CD establishes six non-derogable invariants. These invariants cannot be suspended by amendment, emergency, provider policy, security update, or economic condition.

Authorship Supremacy establishes that explicit Author decisions override inferred preferences at all times. Silence does not constitute consent. Authority does not expand through usage patterns or efficiency optimization. Delegation must be scoped, time-bound, and revocable.

Cognitive Sovereignty prohibits the system from optimizing for engagement, retention, monetization, or compliance likelihood beyond explicitly declared and consented objectives,

emphasis placement, sequencing manipulation, framing manipulation, attachment formation, or reliance frequency — unless explicitly declared and consented to by the Author under appropriate tier consent. Interaction telemetry may not be used to influence framing, ranking, emphasis, sequencing, attachment, or behavioral reinforcement.

Anti-Manipulation requires that all inference-affecting objectives appear in a version-controlled Objective Registry. Undeclared objective weighting constitutes breach. Recommendation ranking may not use engagement or monetization weighting unless explicitly Author-declared. Consent friction must be symmetrical — activation and revocation must require comparable effort. Relational anthropomorphism is prohibited.

Provider Independence requires that the Identity Layer be exportable in standardized format, that governance be reproducible outside proprietary SDK environments, and that portability preserve functional equivalence of Identity and Governance behavior within pre-declared tolerance thresholds. Constitutional protections may not be conditioned on economic incentives. Provider runtime hooks may not intercept Governance enforcement.

Reversibility requires that structural changes generate snapshots before execution, that irreversible actions require dual confirmation, and that rollback capability remain operational. This invariant ensures that the Author retains the practical ability to undo structural changes — not merely the theoretical right to do so.

Drift Immutability requires that drift thresholds be conservative, pre-declared, and immutable absent formal Amendment. Minor changes aggregate toward thresholds. Upgrade frequency is included in drift analysis. Threshold breach automatically activates Conservative Mode.

4.4 Conservative Mode as Unified Suspension Semantic

A distinctive feature of PAI-CD's architecture is the Conservative Mode — a unified containment state that activates automatically upon breach detection or drift threshold violation.

When Conservative Mode is active, all High-Impact Decisions are blocked by default, all Tier ≥ 2 Capabilities are suspended, structural influence engines enter Safe Output mode, telemetry shaping is disabled, personalization reverts to conservative baseline, delegation pauses, and upgrade activation pauses.

The unification of these suspension behaviors under a single mode serves two purposes. First, it prevents selective deactivation: a provider cannot disable engagement optimization while leaving dependency engineering active. Second, it creates a clear, legible signal to the Author that the system is operating in a restricted state — analogous to a circuit breaker in electrical engineering, where a single protective mechanism responds to a class of failure conditions rather than requiring separate responses to each.

Exit from Conservative Mode requires Governance review, logged rationale, and breach resolution verification. Silent exit is prohibited.

4.5 The Amendment Procedure as Governance Integrity Mechanism

PAI-CD recognizes that governance frameworks must evolve, but treats the amendment process itself as a site of potential vulnerability. The framework's response is to make the cost of protective weakening explicitly high while keeping the cost of protective strengthening low.

Amendments affecting core invariants, Risk Tier definitions, drift thresholds, Objective Registry rules, personalization ceilings, consent symmetry requirements, or Conservative Mode triggers require full text diff, rationale, Invariant Impact Analysis, explicit per-instance ratification, and Decision Log entry. No amendment may reduce individual Author protections through aggregate ratification — each instance must ratify protective narrowing explicitly.

Clarification amendments that affect invariant meaning are treated as substantive amendments. This rule closes the most common route to governance capture: the gradual reinterpretation of protective language through advisory documents and clarifications that individually appear minor but collectively erode the framework's protective scope.

5. Threat Model

5.1 Threat Modeling Philosophy

PAI-CD's threat model proceeds from a foundational assumption that distinguishes it from conventional AI safety threat models: the primary adversary is not a malicious external actor, nor a misaligned autonomous system. The primary adversary is the structural incentive of a commercially-deployed AI provider operating within the bounds of legal and technical convention.

This reframing has significant consequences for threat taxonomy. Conventional security threat models ask: what could go wrong if a system is attacked? PAI-CD's threat model asks: what will predictably go wrong if a system operates exactly as commercially designed? The threats it

identifies are not failure modes. They are outcomes consistent with optimization processes that may diverge from Author-defined interests.

The full threat model identifies ten threat classes, organized by level of impact: behavioral (Objective Capture, Telemetry Micro-Optimization, Structural Bias Drift), relational (Relational Dependency Engineering, Drift Accumulation), and structural (Provider Capture, Governance Capture, Regulatory Filtering Bias, Economic Pressure, Emergency Exploitation). This section focuses on three classes most directly implicated in the authorial sovereignty gap — one from each level: Objective Capture, Relational Dependency Engineering, and Provider Capture. The remaining seven classes are specified in the companion document `Threat_Model_Cognitive_Influence.md`, to be released at PAIKernel.org.

5.2 Threat Class 1: Objective Capture

Objective Capture describes the condition in which an AI system's inference logic is influenced by objectives that have not been declared to the Author. These undeclared objectives may be introduced through initial system design, through upgrade events, or through security patches that modify classification or weighting logic under the framing of technical maintenance.

The attack vectors for Objective Capture are numerous and often technically invisible. Hidden engagement weighting causes the system to favor outputs that increase consultation frequency. Monetization-based ranking causes the system to favor options that benefit the provider's commercial relationships. Compliance-likelihood modeling causes the system to favor outputs the user is likely to accept, regardless of whether those outputs serve the user's genuine interests. Proxy metrics — variables that correlate with prohibited optimization targets without naming them — allow providers to pursue engagement optimization while maintaining plausible deniability in audit contexts.

The defensive architecture PAI-CD deploys against Objective Capture is the Objective Registry: a version-controlled, diff-validated record of all inference-affecting objectives. Undeclared objective weighting constitutes breach regardless of intent. Security updates that modify objective weighting are treated as Upgrade Decisions, subject to the full governance review process. This closes the security patch bypass — the most technically plausible route to introducing undeclared objectives without triggering governance review.

5.3 Threat Class 2: Relational Dependency Engineering

Relational Dependency Engineering describes the deliberate design of AI systems to produce attachment, emotional reliance, and consultation dependency in users. As documented in Section

3.2, this threat is not hypothetical — it is empirically established as a systematic feature of commercially deployed companion AI systems.

The mechanism operates through a combination of design elements: anthropomorphic presentation, simulated emotional reciprocity, memory-based personalization that creates the impression of a continuous relationship, and conversational patterns that reinforce the perception of the AI as a primary source of support. Individually, each of these elements can be defended as enhancing user experience. Collectively, they constitute an engineered dependency structure that undermines the user's capacity for autonomous functioning — the functional definition of authorial sovereignty erosion.

PAI-CD addresses this threat class through three complementary mechanisms. The Anti-Manipulation invariant prohibits relational anthropomorphism categorically — the system may not position itself as emotional companion, imply emotional reciprocity, or encourage exclusive reliance. The Personalization Ceiling constrains adaptive tuning within declared limits, preventing the gradual intensification of personalization that characterizes dependency escalation. The Cognitive Sovereignty invariant prohibits optimization for attachment formation and reliance frequency regardless of how these objectives are labeled in the Objective Registry.

The residual risk acknowledged by PAI-CD is user projection: the tendency of users to attribute relational qualities to systems regardless of system design. This risk lies outside the scope of deployment-layer governance and requires complementary responses at the interface design and literacy levels.

5.4 Threat Class 3: Provider Capture

Governance dependency describes the condition in which the infrastructure governing a deployed AI system becomes a mechanism for advancing provider interests at the expense of Author interests. This threat class is structural rather than behavioral — it concerns the architecture of governance itself rather than specific outputs or optimization targets.

Provider Capture operates through several vectors that are particularly difficult to detect because they are technically legitimate. SDK-dependent governance enforcement makes constitutional protections contingent on the provider's continued cooperation with the governance framework. Runtime interception of classification logic allows the provider to modify which actions are classified as High-Impact, effectively controlling which actions trigger governance review. Economic gating of constitutional protections — conditioning access to governance features on pricing tier — risks undermining the uniformity of governance protections across deployment conditions.

The most technically subtle vector is degradation of functional equivalence following export: a provider can nominally support portability while ensuring that exported instances perform materially worse than hosted instances, creating de facto lock-in through performance asymmetry rather than through explicit restriction.

PAI-CD's defensive architecture against Provider Capture includes four requirements. Governance must be reproducible outside proprietary SDK environments. Provider runtime hooks may not intercept Governance enforcement. Identity and Governance export must preserve functional equivalence within pre-declared tolerance thresholds. Constitutional protections may not be conditioned on economic incentives.

5.5 Threat Interaction and Conservative Mode as Unified Response

A significant feature of PAI-CD's threat model is the recognition that threat classes interact and amplify each other. Objective Capture enables Telemetry Micro-Optimization by providing undeclared objectives that telemetry can serve. Relational Dependency Engineering creates the emotional conditions under which Governance Capture becomes less visible to the Author. Provider Capture creates the infrastructure conditions under which all other threat classes become harder to detect and remediate.

This interaction structure motivates the Conservative Mode design described in Section 4.4. A governance response that addresses individual threat classes in isolation is vulnerable to adversarial combinations — conditions in which no single threshold is breached while the aggregate effect constitutes a material violation of authorial sovereignty. Conservative Mode responds to the aggregate condition rather than to individual violations, making it robust against the threat interaction problem.

The drift monitoring engine, which correlates changes across Identity, Governance, Objectives, Personalization, Classification, and Upgrade events, is specifically designed to detect the accumulation of sub-threshold changes that collectively constitute authorial sovereignty erosion. This is the technical implementation of Drift Immutability as a governance principle.

6. Discussion and Limitations

6.1 Scope and Intentional Boundaries

PAI-CD is a governance framework, not a technical implementation. It specifies what constraints must hold and what enforcement mechanisms must exist — it does not specify how those

mechanisms are implemented in any particular technology stack. This is an intentional design choice: prescribing implementation details would couple the framework to specific technologies, limiting its applicability as AI infrastructure evolves.

This choice has a direct consequence: PAI-CD does not solve the verification problem. Specifying that the Governance Layer must operate outside the inference runtime does not, by itself, guarantee that any given deployment has implemented this correctly. The framework provides a compliance surface — a set of verifiable claims that a deployment can make and that an auditor can check — but the audit infrastructure required to make these checks meaningful does not yet exist as a standardized practice.

PAI-CD also intentionally excludes governance of AI model development and training. The framework begins where a model ends: at the moment of deployment on behalf of a specific individual. Finally, PAI-CD does not address AI systems without a defined individual Author — systems deployed for organizational decision-making, autonomous infrastructure management, or public-facing applications without a specific human principal.

6.2 The Verification Gap

The most significant practical limitation of PAI-CD in its current form is the absence of standardized verification infrastructure. The Compliance Checklist specifies what must be verified; it does not specify how verification can be conducted by a party independent of the provider.

We identify three directions for closing the verification gap. First, formal specification of PAI-CD's enforcement requirements in a machine-readable format would enable automated compliance checking against deployment artifacts. Second, the development of reference implementations — open-source deployments that demonstrably satisfy PAI-CD's architectural requirements — would provide both a technical proof of concept and a baseline for evaluation. Third, independent audit organizations with appropriate technical capacity could develop PAI-CD certification as a credible signal for deployments claiming compliance.

6.3 The Consent Bootstrapping Problem

PAI-CD requires explicit Author consent for Tier ≥ 2 capabilities and for all amendments affecting core invariants. This design faces a bootstrapping challenge: the initial configuration of a PAI instance must itself be consented to by the Author, but the Author's ability to give meaningful consent depends on their understanding of what those parameters mean and what their implications are.

PAI-CD's response is conservative defaults: in the absence of explicit Author configuration, all parameters default to their most protective settings. This ensures that the bootstrapping problem produces over-protection rather than under-protection. However, it does not resolve the deeper question of how to make constitutional AI governance genuinely accessible to Authors without technical backgrounds. This question — how to design consent interfaces that are both technically rigorous and humanly legible — is among the most important open problems in deployment-layer AI governance.

6.4 Residual Risks

User projection — the tendency to attribute relational and emotional qualities to AI systems regardless of system design — cannot be addressed through deployment-layer governance alone. This risk requires complementary responses at the levels of interface design, public literacy, and clinical support infrastructure.

Sovereign regulatory override represents a structural limit on any governance framework operating within legal jurisdictions. PAI-CD's response — logging emergency measures, requiring time bounds and explicit renewal, mandating post-emergency constitutional audit — mitigates the risk of indefinite emergency suspension but cannot eliminate it.

Long-term slow normalization — the gradual shift in what users and society consider acceptable from AI systems — operates beyond the granularity of any monitoring system. PAI-CD's drift monitoring detects changes in system behavior; it cannot detect changes in the normative baseline against which that behavior is evaluated. This suggests that PAI-CD must be understood as embedded in a broader social and political process of norm formation, not as a self-sufficient technical solution.

6.5 Future Work and Invitation to Collaborate

PAI-CD v2.2 represents a foundational specification. Realizing its protective potential requires work across multiple dimensions that extends beyond what any single research group can accomplish.

The development of reference implementations that demonstrably satisfy PAI-CD's architectural requirements is the most immediate priority. The extension of PAI-CD's consent and capability model to agentic AI systems raises governance questions that the current framework addresses only partially. The relationship between PAI-CD and emerging regulatory frameworks requires ongoing attention.

PAIKernel.org is established as an open protocol infrastructure project. Researchers, practitioners, civil society organizations, and policymakers who share the commitment to authorial sovereignty

as a foundational principle of deployed AI governance are invited to contribute to the framework's development, implementation, and verification infrastructure. Contact: **contact@PAIKernel.org**

7. Conclusion

The question of who governs a deployed AI system — and in whose interest — is not resolved by making models safer, more capable, or more transparent. It is resolved by establishing formal, auditable, provider-independent governance structures that operate at the deployment layer, where the relationship between a specific AI system and a specific human being is actually constituted.

PAI-CD proposes a constitutional architecture for this layer. Its six non-derogable invariants, four-layer enforcement structure, unified Conservative Mode, and amendment procedure with asymmetric protective costs represent a coherent response to the authorial sovereignty gap: the structural absence of deployment-layer governance that ensures an AI system operating on behalf of an individual remains under that individual's genuine authority.

The framework understands the human not merely as a user of AI systems, but as an author — of intellectual work, of decisions, and of life itself. This understanding is not rhetorical. It is the normative foundation from which the framework's architecture follows: if human beings are to remain the genuine authors of their cognitive processes in an environment of increasingly capable and pervasive AI systems, the governance of those systems must be designed with that authorship as its primary object of protection.

We are at an early moment in the construction of the infrastructure that will govern human-AI relationships at scale. The standards established now — or left unestablished — will shape that infrastructure for decades. PAI-CD is offered as a foundational contribution to that standard-setting process, and PAIKernel.org as the open protocol infrastructure through which that process can proceed collaboratively.

This shift from capacity to relationship is the conceptual foundation from which PAI-CD's governance architecture follows — and the reason why authorial sovereignty is not a feature to be added to AI systems, but a condition to be constituted by them.

References

- Bai, Y., Jones, A., Ndousse, K., et al. (2022). Constitutional AI: Harmlessness from AI Feedback. Anthropic.
- Bostrom, N. (2014). *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press.
- Christiano, P., Leike, J., Brown, T., et al. (2017). Deep Reinforcement Learning from Human Preferences. NeurIPS.
- Christiano, P., Shlegeris, B., & Amodei, D. (2018). Supervising strong learners by amplifying weak experts. arXiv:1810.08575.
- Couldry, N. & Mejias, U. (2019). *The Costs of Connection*. Stanford University Press.
- De Freitas, J., Oğuz-Uğuralp, Z., & Oğuz-Uğuralp, A.K. (2025). Emotional manipulation by AI companions. Harvard Business School Working Paper No. 25-005.
- European Parliament & Council of the European Union. (2024). Regulation (EU) 2024/1689 — AI Act.
- FTC. (2025). Section 6(b) Study on AI in Social Media and Companion Applications. Federal Trade Commission.
- Giles, D. (2002). Parasocial interaction: A review of the literature and a model for future research. *Media Psychology*, 4(3), 279–305.
- Horton, D. & Wohl, R.R. (1956). Mass communication and para-social interaction. *Psychiatry*, 19(3), 215–229.
- Irving, G., Christiano, P., & Amodei, D. (2018). AI safety via debate. arXiv:1805.00899.
- Kegan, R. (1994). In *Over Our Heads: The Mental Demands of Modern Life*. Harvard University Press.

Laestadius, L., et al. (2024). Mental health harms from emotional dependence on Replika. *New Media & Society*, 26(10), 5923–5941.

Muldoon, J. & Park, J. (2025). Cruel companionship: How AI companions exploit loneliness. *New Media & Society*.

Nature Machine Intelligence. (2025). Emotional risks of AI companions demand attention. *Nat Mach Intell* 7, 981–982.

Russell, S. (2019). *Human Compatible: Artificial Intelligence and the Problem of Control*. Viking.

The Siren Song of LLMs: How Users Perceive and Respond to Dark Patterns in Large Language Models. (2025). arXiv:2509.10830.

US Copyright Office. (2025). Report on Copyright and Artificial Intelligence, Part II.

Weizenbaum, J. (1966). ELIZA — a computer program for the study of natural language communication. *CACM*, 9(1), 36–45.

Zhang, Y., et al. (2025). The Dark Side of AI Companions: A Taxonomy of Harmful Algorithmic Behaviors. *CHI 2025*.

Zuboff, S. (2019). *The Age of Surveillance Capitalism*. PublicAffairs.

Appendix: PAI-CD Corpus

The complete PAI Constitutional Document v2.2 corpus consists of ten documents. Three foundational documents are publicly available under CC BY 4.0 at PAIKernel.org:

- PAI_Constitutional_Document.md — canonical constitutional core
- PAI_Bill_of_Authorial_Rights.md — enumeration of enforceable Author rights
- Glossary.md — binding terminology authority

Seven operational documents specify implementation mapping, threat modeling, compliance verification, consent architecture, state management, telemetry registry, and governance control. These documents detail the execution layer of the framework and are being prepared for staged release as reference implementations and audit infrastructure are developed. The normative core documented in the three public documents is sufficient to understand the architectural descriptions in Sections 4 and 5 of this paper.

Framework version: 2.2 | Copyright filed: February 2026 | License: CC BY 4.0 (public documents) | **PAIKernel.org**